UNIT 7
NETWORK SECURITY

LH - 5HRS

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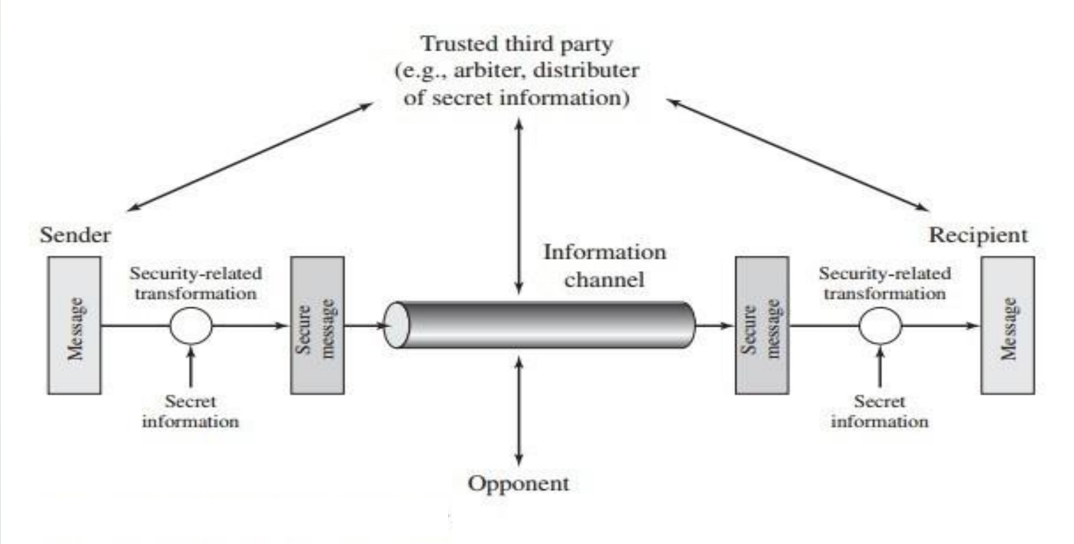
7.1 A Model for Network Security

A general model for network security is shown in figure below:



Security aspects come into play when it is necessary to protect the information transmission from an opponent who may present a threat to confidentiality, integrity and so on.

All the techniques for providing security have two components:

- A security-related transformation on the information to be sent. Examples include the encryption of the message, which scrambles the message so that it is unreadable by the opponent, and the addition of a code based on the contents of the message, which can be used to verify the identity of the sender.
- Some secret information shared by the two principals and, it is hoped, unknown to the opponent. An example is an encryption key used in conjunction with the transformation to scramble the message before transmission and unscramble it on reception.

- A trusted third party (TTP) may be needed to achieve secure transmission. For example, a third party may be responsible for distributing the secret information to the two principals while keeping it from any opponent. Or a third party may be needed to arbitrate disputes between the two principals concerning the authenticity of a message transmission.

This general model shows that there are four basic tasks in designing a particular security service:

(1) <u>Design an algorithm</u> for performing the security-related transformation. The algorithm should be such that an opponent cannot defeat its purpose.

(2) <u>Generate the secret information</u> to be used with the algorithm.

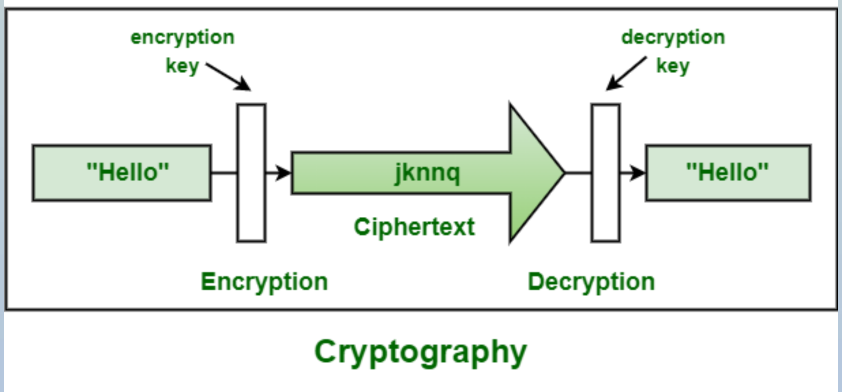
(3) <u>Develop methods for the distribution</u> and sharing of the secret information.

(4) <u>Specify a protocol</u> to be used by the two principals that makes use of the security algorithm and the secret information to achieve a particular security service.

7.2 Principles of Cryptography: Symmetric Key and Public Key

Cryptography is technique of securing information and communications through use of codes so that only those person for whom the information is intended can understand it and process it. Thus preventing unauthorized access to information. The prefix “crypt” means “hidden” and suffix graphy means “writing”.

In today’s age of computers cryptography is often associated with the process where an ordinary plain text is converted to cipher text which is the text made such that intended receiver of the text can only decode it and hence this process is known as encryption. The process of conversion of cipher text to plain text this is known as decryption.



Features Of Cryptography are as follows:

1. Confidentiality:

Information can only be accessed by the person for whom it is intended and no other person except him can access it.

2. Integrity:

Information cannot be modified in storage or transition between sender and intended receiver without any addition to information being detected.

3. Non-repudiation:

The creator/sender of information cannot deny his or her intention to send information at later stage.

4. Authentication:

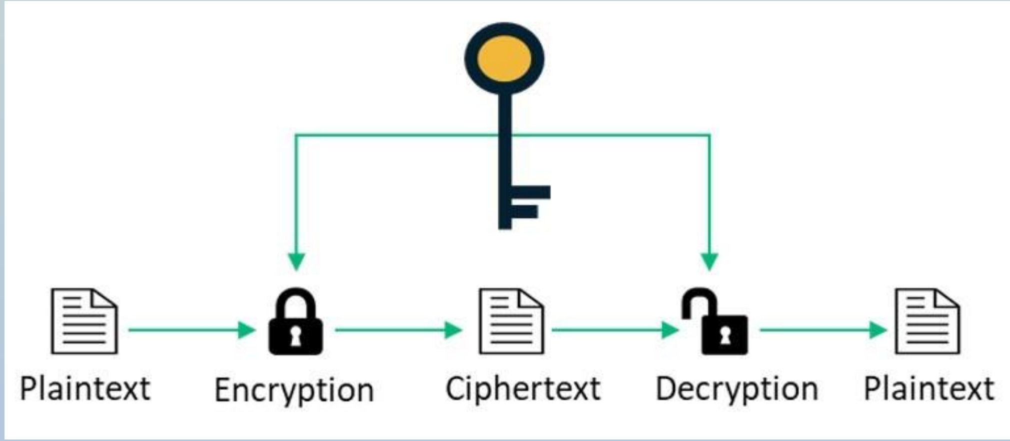
The identities of sender and receiver are confirmed. As well as destination/origin of information is confirmed.

A basic cryptosystem includes the following components:

- Plaintext- This is the data that needs to be protected.
- Encryption algorithm- This is the mathematical algorithm that takes plaintext as the input and returns ciphertext. It also produces the unique encryption key for that text.
- Ciphertext- This is the encrypted, or unreadable, version of the plaintext.
- Decryption algorithm- This is the mathematical algorithm that takes ciphertext as the input and decodes it into plaintext. It also uses the unique decryption key for that text.
- Encryption key- This is the value known to the sender that is used to compute the ciphertext for the given plaintext.
- Decryption key- This is the value known to the receiver that is used to decode the given ciphertext into plaintext.

1. Symmetric-key algorithm (secret-key cryptography or private-key encryption):

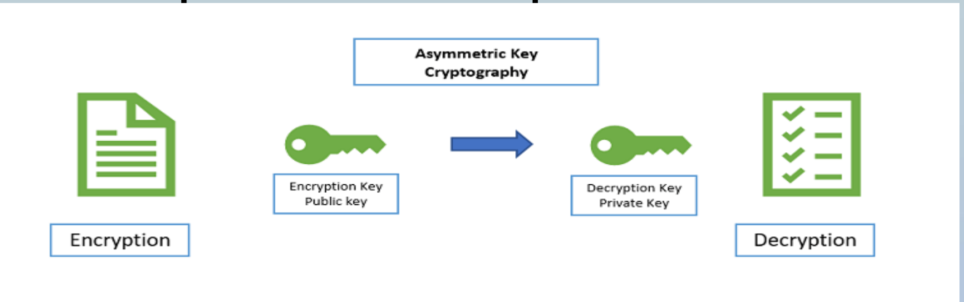
- These algorithms use the same pre-shared key, sometimes called a secret key pair, to encrypt and decrypt data.
- Both the sender and receiver know the pre-shared key before any encrypted communication begins.
- As shown in figure below, symmetric algorithms use the same key to encrypt and decrypt the plaintext.



- For example, Alice and Bob live in different locations and want to exchange secret messages with one another over an insecure channel. Alice encrypts the message using an encryption algorithm; Bob decrypts the message using a decryption algorithm. It uses a single secret key for both encryption and decryption.
- If Bob wants to talk to Eve, he needs a new pre-shared key for the communication to keep it secret from Alice. The more people Bob wants to communicate securely, the more keys he will need to manage.
- Some examples of symmetric encryption algorithms include:
  - AES (Advanced Encryption Standard)
  - DES (Data Encryption Standard)

2. Asymmetric-key algorithm (Public-key encryption or public-key cryptography):

- Asymmetric cryptography is a branch of cryptography where a secret key can be divided into two parts, a public key and a private key.
- The public key can be given to anyone, trusted or not, while the private key must be kept secret (just like the key in symmetric cryptography).
- Any person can encrypt a message using the public key of the receiver, and the receiver is the only one that can decrypt it using his private key.
- Asymmetric algorithms are more complex and requires intensive resource and slower to execute.



- For example, Alice and Bob live in different locations and want to exchange secret messages with each other over an insecure channel.
- To send a secured message to Bob, Alice or anyone encrypts the message using Bob’s’ public key.
- To decrypt the message, Bob uses his own Private key.
- Some examples of asymmetric encryption algorithms include:
  - Rivest-Shamir-Adleman (RSA)
  - Digital Signature Algorithm (DSA)

7.3 Public Key Algorithm- RSA

- RSA(Rivest-Shamir-Adleman) algorithm is asymmetric cryptography algorithm and is considered as the most secure way of encryption.
- Asymmetric actually means that it works on two different keys i.e. Public Key and Private Key. Public Key is given to everyone and Private key is kept private.

An example of asymmetric cryptography :

- A client (for example browser) sends its public key to the server and requests for some data.
- The server encrypts the data using client's public key and sends the encrypted data.
- Client receives this data and decrypts it.
- Since this is asymmetric, nobody else except browser can decrypt the data even if a third party has public key of browser.

Working/Steps of RSA(Rivest–Shamir–Adleman) Algorithm:

Step 1: Choosing two different large prime numbers (say p & q).

Step 2: Calculating $n=p*q$.

Step 3: Calculating $\phi(n)=(p-1)(q-1)$.

Step 4: Choosing public key (e) such that $1<e<\phi(n)$; e is co-prime to $\phi(n)$.

Step 5: Calculating private key (d) such that $de \equiv 1 \mod \phi(n)$ or $de=1+k.\phi(n)$ where $k=0,1,2,\ldots$.

Question:

In a RSA cryptosystem, A uses two prime numbers p=13 and q=17 to generate his public and private keys. If public key is 35, then private key is _____________.

a) 11 b) 13 c) 16 d) 17

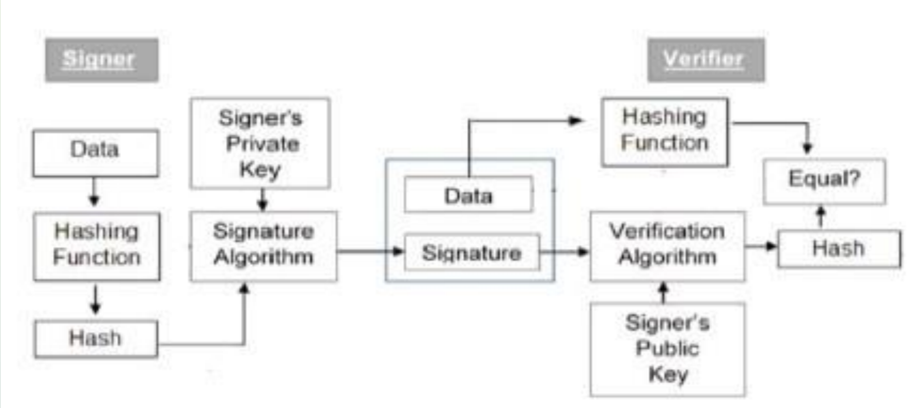
7.4 Digital Signature Algorithm (DSA)

- It is a mathematical technique used to validate the authenticity and integrity of a message.
- As a digital equivalent of a handwritten signature, a digital signature offers far more inherent security and it is intended to solve the problem of tampering and impersonation in digital communications.
- Digital signatures can provide the added assurances of evidence of origin, identity and status of an electronic document, transaction or message.

Digital signatures must have the following properties:

- **Authentication**: A digital signature must give the receiver reason to believe the message was created and send by the claimed sender.
- **Non-repudiation**: With digital signature, the sender can’t deny having sent the message later on.
- **Integrity**: A digital signature must ensure that the message was not altered in transit.

Digital Signature Process



- Figure above shows the digital signature process.
- The sender uses a signing/signature algorithm to sign the message (signature=only hash of the message encrypted using sender's private key).
- The message and signature are sent to the receiver. The receiver receives the message and the signature and applies the verifying algorithm to the combination.
- If the result is true, the message is accepted; otherwise, it is rejected.

Advantages of the Digital Signature Algorithm

- Along with having strong strength levels, the signature’s length is smaller compared to other digital signature standards.
- DSA requires less storage to work as compared to other digital standards.
- DSA is patent-free so that it can be used free of cost.

Disadvantages of Digital Signature Algorithm

- It requires a lot of time to authenticate as the verification process includes complicated remainder operators. It requires a lot of time for computation.
- Data in DSA is not encrypted. We can only authenticate data in this.
- The digital signature algorithm firstly computes with SHA1 hash and signs it. Any drawbacks in the cryptographic security of SHA1 are reflected in DSA because implicitly of DSA is dependent on it.

7.5 Communication Security

1. IPSec:

The IP security (IPSec) is an Internet Engineering Task Force (IETF) standard suite of protocols that operates at the network layer (Layer 3) to authenticate and encrypt each IP packet in a communication session, securing data as it travels across networks.

It uses **AH (Authentication Header)** for data integrity and origin verification, and **ESP (Encapsulating Security Payload)** for encryption + integrity, working in either **Transport mode** (end-to-end) or **Tunnel mode** (gateway-to-gateway).

It uses **IKE (Internet Key Exchange)** to automatically negotiate and manage cryptographic keys and security associations (SAs) between two communicating parties.

Uses of IP Security -

Use Case	How IPSec is Used
VPNs	Most corporate VPNs (site-to-site & remote access) run on IPSec to create encrypted tunnels
Remote Work	Employees connect securely to office networks over public internet
Branch Office Connectivity	Two office locations linked securely via IPSec tunnel between routers
Cloud Connectivity	Secure connection between on-premise data centers and AWS/Azure/GCP
IoT & Critical Infrastructure	Securing communication between devices in power grids, factories

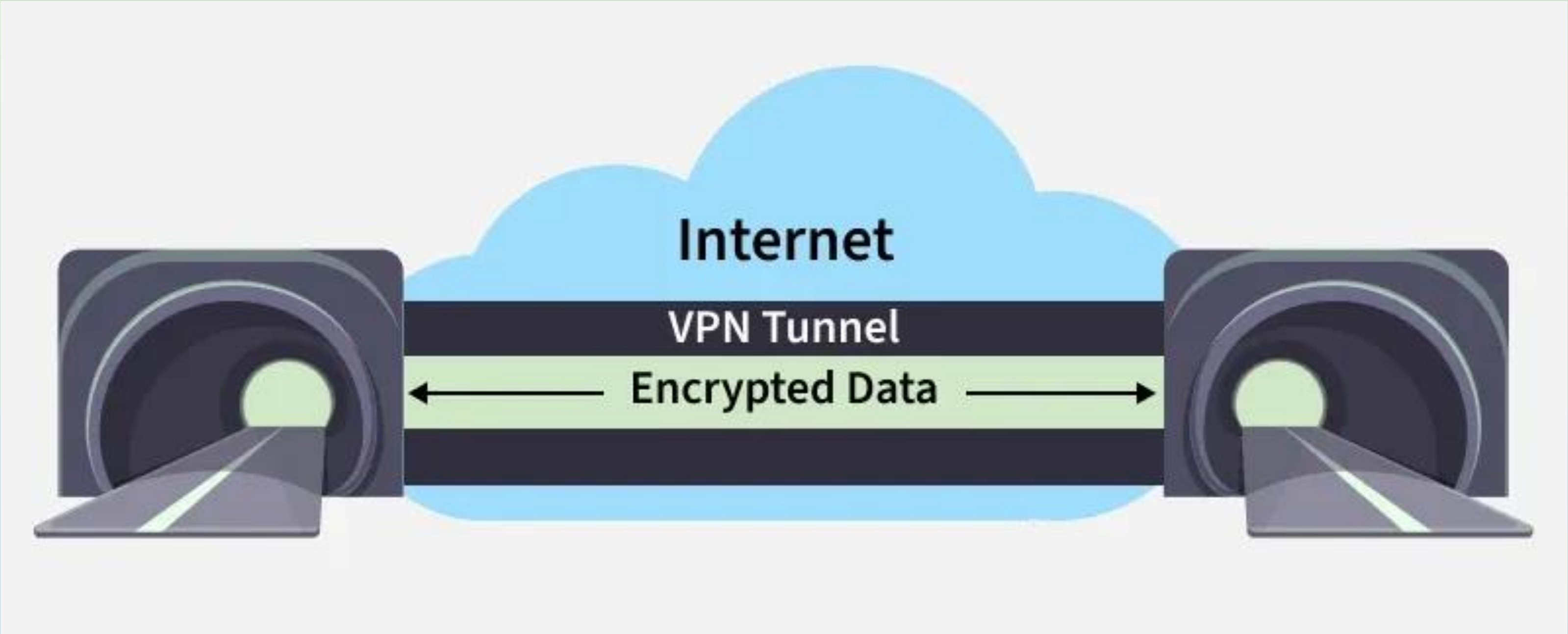
Virtual Private Network VPN

A VPN is a **technology that creates a secure, encrypted tunnel over a public network (like the internet)**, allowing users or networks to communicate privately as if they were directly connected on a private network.

An employee can work outside the office and still securely connect to the corporate network

Traffic on the virtual network is sent securely by establishing an encrypted connection across the Internet known as a tunnel. VPN traffic from a device such as a computer, tablet, or smartphone is encrypted as it travels through this tunnel.

Virtual Private Network VPN



Types of VPN

1. **Remote Access VPN:** It allows an individual user to securely connect to a private network over the internet, and it is widely used by employees working remotely.

2. **Site-to-Site VPN:** It securely connects two or more separate networks, such as a head office and branch offices, so internal communication remains protected across locations.

Advantages of VPN:

1. VPNs are inexpensive

VPN services are generally affordable compared to dedicated private network connections. Many VPN providers offer low-cost monthly or yearly subscriptions, making them accessible to students, businesses, and individual users. Organizations can securely connect remote employees without investing heavily in expensive networking infrastructure.

2. Helps in bypassing blockers and filters

A VPN hides the user’s real IP address and routes internet traffic through secure remote servers. Because of this, users can bypass:

- Geographic restrictions
- Internet censorship
- Website blocking
- Network filters imposed by schools or organizations

For example, if a website is blocked in a country or institution, users can connect through a VPN server located in another region to access it.

Advantages of VPN:

3. Provides enhanced security

VPNs improve security by encrypting internet traffic. Encryption converts readable data into unreadable form, protecting sensitive information from:

Hackers

Cybercriminals

Eavesdroppers

Unauthorized monitoring

This is especially important when using public Wi-Fi networks in cafes, airports, or hotels.

4. Allows a strong degree of online anonymity

VPNs mask the user’s original IP address and location, making online activities more private. Websites and trackers cannot easily identify the real user location or browsing behavior. This helps protect user privacy and reduces online tracking by advertisers and third parties

3. Firewalls:

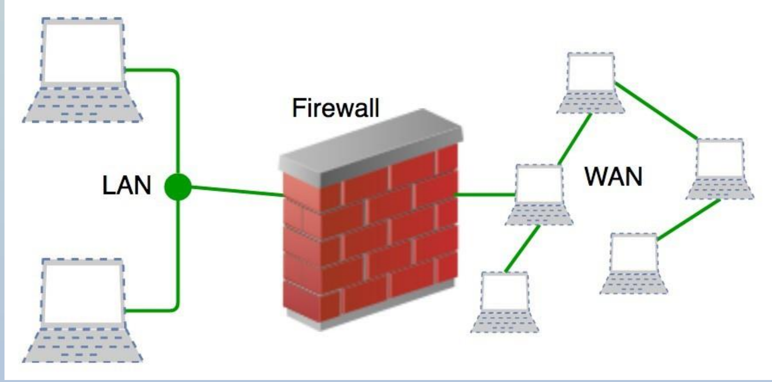
- A firewall is a network security device, either hardware or software-based, which monitors all incoming and outgoing traffic and based on a defined set of security rules it accepts, rejects or drops that specific traffic.

Accept : allow the traffic

Reject : block the traffic but reply with an “unreachable error”

Drop : block the traffic with no reply

- A firewall establishes a barrier between secured internal networks and outside untrusted network, such as the Internet.



Advantages:

- They can stop incoming suspicious requests.
- They are more cost effective.

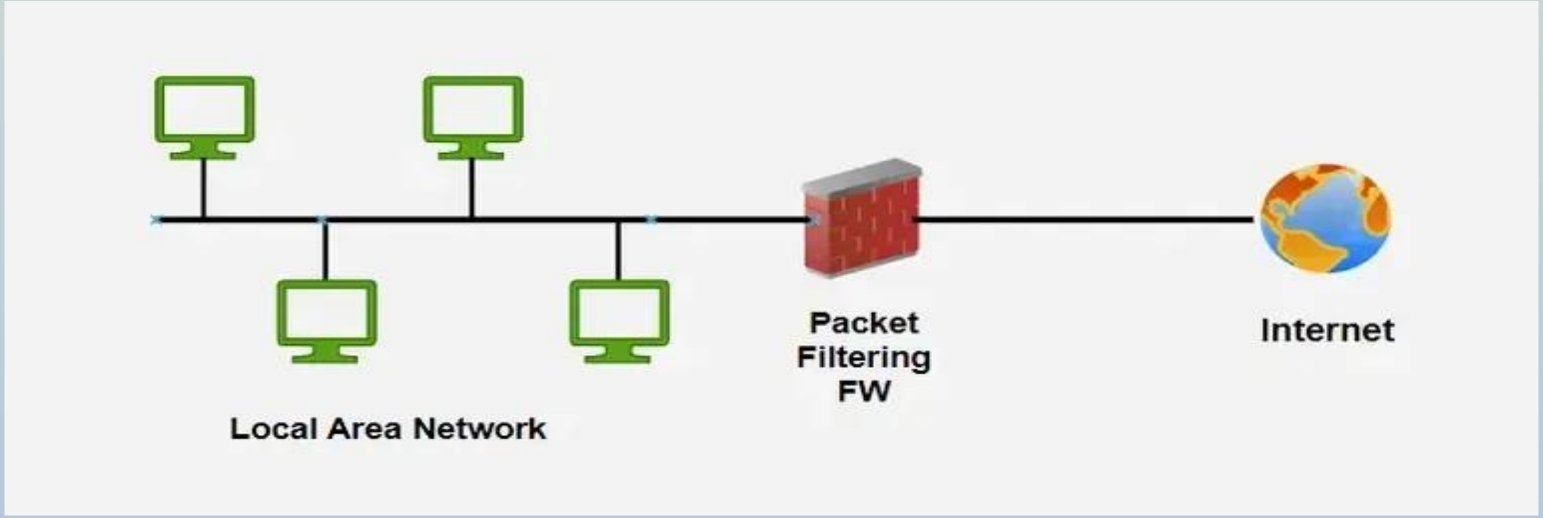
Disadvantages:

- They are a central point for attack.
- They do not protect against back door attacks.

1. Packet Filtering Firewall

A basic firewall that checks packet headers like IP, port, and protocol.

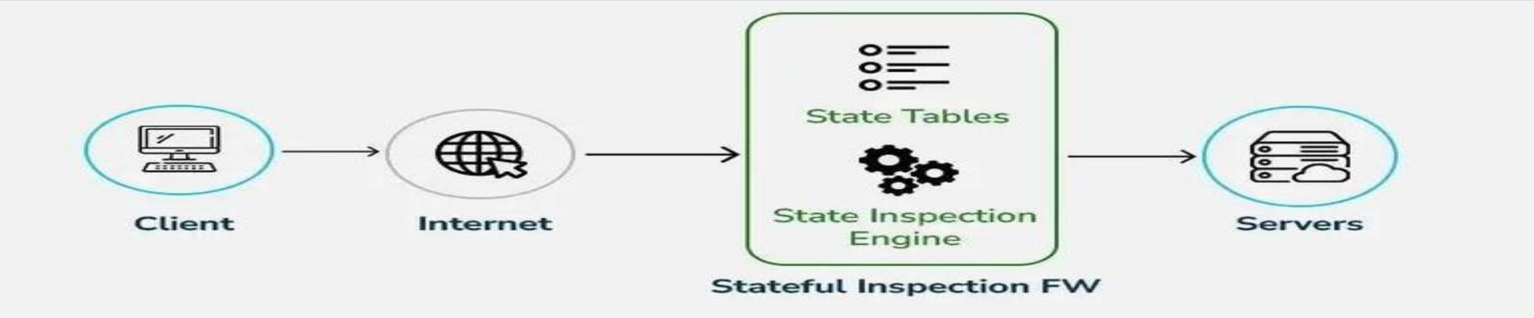
- Very fast and lightweight
- Does not inspect data inside packets
- Provides basic security only



2. Stateful Inspection Firewall

Tracks active connections and makes decisions based on traffic context.

- More secure than packet filtering
- Remembers past traffic (state table)
- Blocks suspicious or unexpected packets



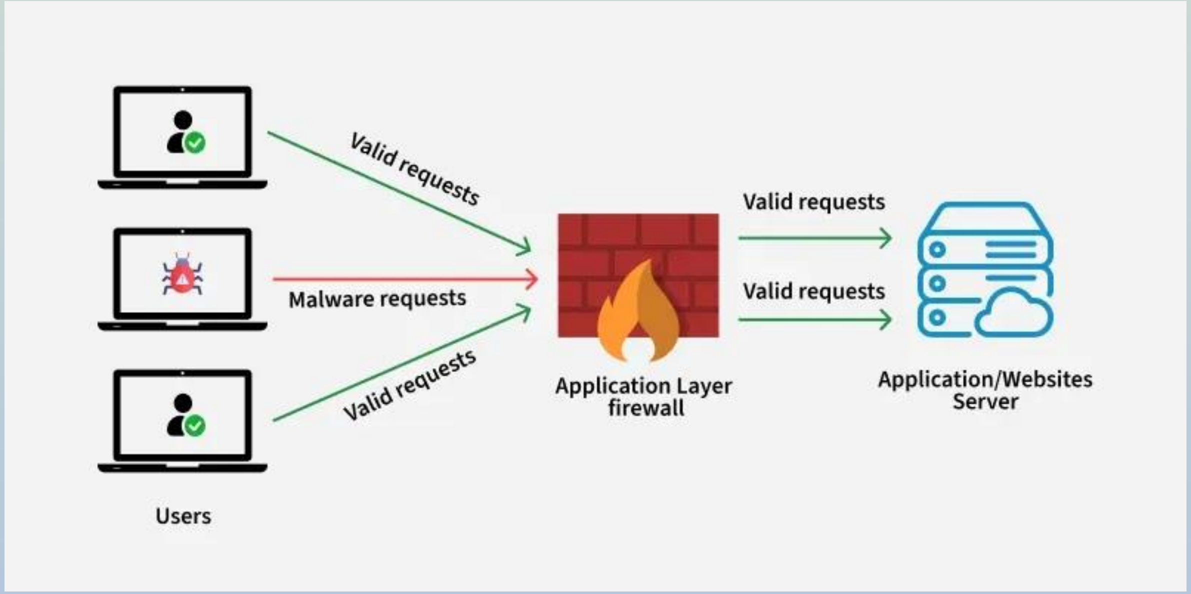
3. Proxy (Application-Level) Firewall

Acts as a middleman between user and destination server.

•Filters data at the application layer

•Hides internal network details

•Can block malicious content before it reaches the user



4. Wireless Security:

Wireless networks provide various comforts to end users, but actually, they are very complex in their operation.

Many protocols and technologies are working behind the scenes to provide a stable connection to users.

Data packets traveling over a wire provide a sense of security to users, as they are probably not overheard by eavesdroppers.

To secure the wireless connection, we should focus on the following areas

Authentication: Identify the endpoint of the wireless network and end-users

Privacy: Protecting wireless data packets from a middleman.

Integrity: Keeping the wireless data packets intact

1. Wired Equivalent Privacy (WEP)

In wireless communication, **open authentication offers no security**. **WEP** uses the **RC4 cipher** to encrypt and decrypt data with a shared **WEP key**, which can serve as both authentication and encryption.

A client can connect to an AP only if it has the correct key. The AP verifies this by sending a challenge phrase; if the client encrypts it correctly, access is granted.

2. Wi-Fi Protected Access (WPA) Protocol:

- WPA arrived as WEP's substitute due to the vulnerabilities contained within WEP. It has additional features, such as the Temporary Key Integrity Protocol (TKIP)-encryption method. This function was a 128-bit dynamic key that was harder to break into than a WEP static, unchanged key. WPA was a major improvement over WEP, but as the core components were rendered so that they could be rolled out through firmware updates to WEP-enabled devices, they still relied on exploited elements.

3. Wi-Fi Protected Access 2 (WPA2) Protocol:

- WPA2 is the successor to WPA and WPA2 has stronger security and is easier to configure than the prior options. The main difference with WPA2 is that it uses the Advanced Encryption Standard (AES) instead of TKIP.
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4. Wi-Fi Protected Access 3 (WPA3) Protocol:

- WPA3 is the new new version of WPA, and one can find it in the routers that were created in 2019. With this new format, WPA3 introduces stronger security to public networks to prevent hackers from extracting information from them.